Inspection Performance Analysis Dashboard

Portfolio Case Study | Operational Reporting with Microsoft Excel and Power BI

DATA NOTE
This project uses simulated data created for learning and portfolio demonstration. It does not represent official records or the actual performance of any organization.

1. Project objective

Create a performance dashboard to track inspection violations, compare total and processed cases by month, and summarize processing performance through key performance indicators (KPIs).

2. Key performance indicators

133	104	78.2%	29
Total violations	Processed	Processing rate	Not processed

3. Tools and workflow

Tools: Microsoft Excel, Power BI, and DAX for KPI measures.
Workflow: organize monthly data in Excel; import and review it in Power BI; create KPI measures and visualizations; compare monthly totals with processed cases.

4. KPI definitions

Processing rate = processed violations ÷ total violations × 100.
Not processed = total violations – processed violations.

Results and Interpretation

The simulated dataset contains 133 violations across four months. Of these, 104 are marked as processed and 29 remain not processed. The overall processing rate is 78.2%.

5. Monthly results

Month	Total violations	Processed	Processing rate
January	30	23	76.7%
February	35	27	77.1%
March	31	24	77.4%
April	37	30	81.1%
Total	**133**	**104**	**78.2%**

6. Key observations

• April has the highest monthly processing rate in this sample (81.1%); January has the lowest (76.7%).
• In each month, processed cases are fewer than total cases, so some workload remains outstanding.
• The data describe this simulated sample only; they do not explain why performance differs or establish a real-world trend.

7. Recommendations for further analysis

1. Add case age or days outstanding to identify overdue cases.
2. Add inspection area or violation category to understand where cases occur most often.
3. Define a target processing rate and compare actual performance with the target.
4. Use verified and authorized operational data before making real-world decisions.

Skills demonstrated

Data organization • KPI calculation • Monthly comparison • Data visualization • Operational reporting • Communicating findings and limitations

This report accompanies the Inspection Performance Analysis Dashboard in the portfolio. All values and findings are based on simulated training data.